

## C.4 Standing waves and resonance

Topic warm-up · IB Physics (2023 syllabus)

with worked answers

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Total: **26 marks**

Speed of sound in air =  $340 \text{ m s}^{-1}$ . Show your working.

### Question 1 — The garage band

(9 marks)

A guitarist tunes the A-string of her guitar. The vibrating length of the string is 0.65 m, fixed at both ends, and it sounds its fundamental at 110 Hz.

- (a) Explain how a standing wave forms on the string, and why the two fixed ends must be nodes. [3]
- (b) Write down the wavelength of the fundamental. [1]
- (c) Calculate the speed of the wave on the string. [2]
- (d) The guitarist lightly touches the string so it vibrates in its third harmonic. State the frequency heard, and the number of nodes and antinodes on the string. [3]

### Question 2 — The milk-bottle band

(10 marks)

For the talent show, a group tunes glass bottles by pouring water into them. Blowing across the top of a bottle makes the air column resonate as a pipe *closed at the bottom* and *open at the top*. One bottle's air column is 0.24 m long.

- (a) State what is found at each end of the resonating air column (node or antinode), and why. [2]
- (b) Calculate the fundamental frequency of this bottle. [2]
- (c) A bandmate predicts the next resonance at about 710 Hz, “double the fundamental”. Show that the next resonance is in fact near 1.1 kHz, and explain the bandmate's error. [3]
- (d) A cardboard tube *open at both ends* is cut to the same 0.24 m. Calculate its fundamental frequency, and state why it differs from the bottle's. [3]

### Question 3 — Chocolate physics

(7 marks)

With the turntable removed, a bar of chocolate left in a microwave oven melts first at a row of evenly spaced spots. The spots are the antinodes of a standing electromagnetic wave, and they are found to be 6.0 cm apart.

- (a) State two differences between a standing wave and a travelling wave. [2]
- (b) Explain why the melted spots are half a wavelength apart, not a full wavelength. [2]
- (c) Hence calculate the frequency of the oven's microwaves ( $c = 3.0 \times 10^8 \text{ m s}^{-1}$ ). [3]

## Answers

### Question 1 — The garage band

- (a) The travelling wave reflects at each fixed end, and the incident and reflected waves, having the same frequency and amplitude but opposite directions, superpose. At positions where they are permanently in antiphase the displacement is always zero, giving nodes. The fixed ends cannot move at all, so they are forced to be nodes; this is what selects the allowed wavelengths.
- (b)  $\lambda_1 = 2L = 2(0.65) = 1.30 \text{ m}$ .
- (c)  $v = f\lambda = 110 \times 1.30 = 143 \text{ m s}^{-1}$  (about  $140 \text{ m s}^{-1}$ ).
- (d)  $f_3 = 3f_1 = 330 \text{ Hz}$ . The third harmonic has three loops, so there are **4 nodes** (counting both fixed ends) and **3 antinodes**.

### Question 2 — The milk-bottle band

- (a) At the closed end (the water surface) there is a *node*: the air cannot move there. At the open top there is an *antinode*: the air is free to oscillate.
- (b) A closed pipe fits a quarter wavelength in its fundamental, so  $\lambda_1 = 4L = 4(0.24) = 0.96 \text{ m}$  and  $f_1 = v/\lambda_1 = 340/0.96 = 354 \text{ Hz}$  (about  $350 \text{ Hz}$ ).
- (c) A pipe closed at one end supports *odd* harmonics only, so the next resonance is  $3f_1 = 3(354) = 1.06 \text{ kHz}$ , near  $1.1 \text{ kHz}$ . The bandmate applied the string and open-pipe rule, where every whole-number multiple is allowed, to a closed pipe where the even ones are forbidden.
- (d) Open at both ends, both ends are antinodes, so the fundamental fits *half* a wavelength rather than a quarter:  $\lambda_1 = 2L = 0.48 \text{ m}$  and  $f_1 = 340/0.48 = 708 \text{ Hz}$ . That is double the bottle's fundamental, because the same length now accommodates twice as much of a wave. (Note this is the very frequency the bandmate predicted in (c): their instinct was right for the wrong pipe.)

### Question 3 — Chocolate physics

- (a) Any two of: a standing wave transfers no net energy along its length, whereas a travelling wave transports energy; in a standing wave the amplitude depends on position, from zero at nodes to maximum at antinodes, whereas every point of a travelling wave oscillates with the same amplitude; in a standing wave all points between adjacent nodes move in phase, whereas in a travelling wave the phase advances steadily along the direction of travel.
- (b) Antinodes, like nodes, repeat every *half* wavelength. Adjacent antinodes are therefore  $\lambda/2$  apart, not  $\lambda$ .
- (c)  $\lambda = 2 \times 0.060 = 0.12 \text{ m}$ , so  $f = c/\lambda = 3.0 \times 10^8/0.12 = 2.5 \times 10^9 \text{ Hz} = 2.5 \text{ GHz}$ .